

A Comparative Analysis of StandardScaler and MinMaxScaler Normalization for Deep Neural Network-Based Dengue Fever Classification

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

Abstract—Dengue Hemorrhagic Fever (DHF) remains a major public health problem in tropical and subtropical regions, making rapid and accurate diagnosis crucial to reduce complications and mortality. Clinical datasets used for DHF classification contain various numerical features with varying value ranges, which can impact the performance of deep learning models. Although data normalization has been widely used in preprocessing, studies comparing the effectiveness of various normalization techniques for DHF classification are limited. This study analyzes the effect of three normalization approaches—no normalization, StandardScaler, and MinMaxScaler—on the performance of a Deep Neural Network (DNN) model using two dataset configurations: a complete dataset containing clinical symptoms and laboratory test results, and a dataset containing only clinical symptoms for an initial screening scenario. Model performance was evaluated using Stratified 5-Fold Cross Validation, and the final model was tested on a separate test dataset. The results showed that data normalization significantly improved classification performance. Among the evaluated methods, StandardScaler performed best on the full dataset with an accuracy of 96.96%, an F1-score of 97.00%, and an ROC-AUC of 0.9972. These findings suggest that selecting the right normalization technique can improve the accuracy, reliability, and stability of deep learning-based dengue fever classification models in processing heterogeneous clinical data.

Index Terms—Dengue Hemorrhagic Fever, Deep Neural Network, Data Normalization, StandardScaler, Clinical Data Classification

I. INTRODUCTION

Dengue Hemorrhagic Fever (DHF) remains a major health challenge, particularly for people in tropical and subtropical countries. This disease is transmitted through the bite of the *Aedes aegypti* mosquito, and the number of cases has continued to increase over the past two decades. According to the World Health Organization (WHO), an estimated 100–400 million cases of dengue fever occur annually worldwide, with Southeast Asia and Latin America being the most affected regions. [1]. Given that this disease can progress rapidly and cause serious complications, early and accurate diagnosis is crucial for improving patient success and reducing mortality. [2], [3]

In clinical practice, the diagnosis of dengue fever is generally based on a combination of patient symptoms and

laboratory findings, such as platelet count, leukocyte count, hematocrit, and hemoglobin levels [4]. However, these clinical manifestations often overlap with other febrile illnesses, making diagnosis very difficult in the early stages of infection. To support clinical decision-making, Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL), are increasingly being adopted for medical classification tasks due to their ability to identify complex relationships in healthcare data [5], [6]. Among these approaches, Deep Neural Networks (DNNs) have demonstrated strong performance in classifying structured clinical data [7]–[9].

Despite these advances, the effectiveness of deep learning models is significantly affected by the quality of data preprocessing. Clinical datasets typically include heterogeneous numeric features with varying value ranges, which can slow model convergence and reduce predictive performance if not scaled [10]. Although feature normalization has been widely incorporated into previous dengue classification studies, relatively few have systematically compared different normalization techniques under multiple dataset configurations. To address this gap, this study evaluates three normalization strategies—no normalization, *StandardScaler*, and *MinMaxScaler*—using DNNs trained on two dataset configurations: a full dataset containing clinical symptoms and laboratory results, and a symptom-only dataset. Model robustness and generalization were assessed using five-fold cross-validation, while classification performance was measured using Accuracy, Precision, Recall, F1-Score, and ROC-AUC [11].

II. RELATED WORK

Previous research has shown that both Machine Learning (ML) and Deep Learning (DL) are effective approaches for dengue classification. A systematic review by Andrade Girón et al. [12] highlighted the successful application of algorithms such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). The review also emphasized that data preprocessing, particularly feature normalization and K-Fold Cross Validation, plays a key role in improving classification performance

and model generalization. Similarly, previous research has shown that normalization methods, including Z-score and Min-Max Scaling, along with cross validation, contribute to higher classification accuracy and more reliable model performance [12]–[16].

Several dengue classification studies have reported accuracy above 97% using ANN and SVM models with appropriate preprocessing and validation strategies [17]–[20]. Other comparative studies achieved accuracy exceeding 99% using feature normalization combined with machine learning algorithms such as Extra Tree Classifier and Decision Tree [3], [21]. Similar findings have also been reported in Indonesia, where Min-Max Scaling improved the performance of an XGBoost-based dengue classification model [5]. Overall, these studies show that feature scaling and validation strategies substantially impact model accuracy and generalization [22].

Although previous research has demonstrated the benefits of normalization, most studies have only evaluated one preprocessing technique or one dataset configuration. Comprehensive comparisons of different normalization methods for DNN-based dengue classification across different dataset configurations are rare. To address this limitation, this study systematically compares three normalization strategies using complete clinical-laboratory data and symptom-only datasets.

III. RESEARCH METHODOLOGY

A. Research Workflow

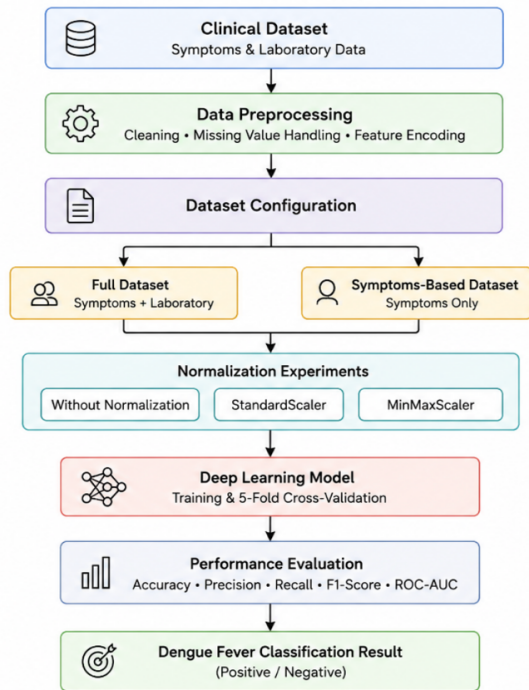


Fig. 1. Research Workflow

Fig. 1 presents the overall workflow of this study. Clinical symptom and laboratory data were first collected and pre-processed through data cleaning, missing value handling, and

feature encoding. The processed data were then divided into two dataset configurations: a complete dataset containing both clinical symptoms and laboratory results, and a symptoms-only dataset representing early-stage screening scenarios. Each dataset configuration was evaluated using three normalization strategies: no normalization, Standard Scaling, and Min-Max Scaling. The normalized data were subsequently used to train a Deep Neural Network (DNN) with five-fold cross-validation. Finally, model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to determine the most effective normalization strategy for dengue fever classification.

B. Dataset

The dataset used in this study consists of clinical records collected from patients presenting with febrile illness. Each record includes demographic information, vital signs, clinical symptoms, laboratory examination results, and a diagnosis label indicating either Dengue Fever or non-dengue fever. In total, the dataset contains 21 input features and one target label, as summarized in Tables I and II.

TABLE I
INPUT FEATURE CATEGORIES

Category	Attributes	Experiments Applied
Demographic Data	Age, Gender	Age Only
Vital Signs	Temperature, Systolic Blood Pressure, Diastolic Blood Pressure	Yes
Laboratory Features	Platelet, Leukocyte, Hematocrit, Hemoglobin	Yes
Clinical Symptoms	Fever, Fever Duration, Headache, Nausea, Vomiting, Abdominal Pain, Fatigue, Loss of Appetite, Red Spots, Body Pain, Abdominal Distension, Joint Pain	Fever Duration Only

TABLE II
TARGET LABEL DESCRIPTION

Target Label	Description
0	Fever
1	Positive Dengue Fever

This dataset consists of 1,141 patient medical records collected from Permata Gading Clinic between January 2024 and December 2025. The class distribution is relatively balanced, consisting of 575 non-dengue cases and 566 dengue cases; therefore, no additional resampling techniques, such as SMOTE, were applied. Normalization was performed only on continuous numeric features, including age, body temperature, systolic and diastolic blood pressure, laboratory measurements, and fever duration. Binary variables representing clinical symptoms, gender, and target label were excluded from the normalization process because their values already represent categorical information.

C. Data Preprocessing

Data preprocessing was performed to improve data quality before model training. Duplicate data was removed, while missing values for numeric and categorical attributes were filled using the median input method. Next, categorical variables, such as gender, clinical symptoms, and diagnostic labels, were converted to binary values for processing by the Deep Neural Network (DNN) model. All 21 features were retained due to their clinical relevance to dengue diagnosis.

After preprocessing, the dataset was split into training-validation and testing data with an 80:20 ratio using stratified sampling. The training process was performed using five-fold stratified cross-validation to evaluate the model's stability and generalization ability. Next, the final model was retrained using the entire training-validation data with an 85:15 internal split for early stopping. It was then evaluated on a held-out test set to obtain more objective results and minimize selection bias.

D. Normalization Experiments

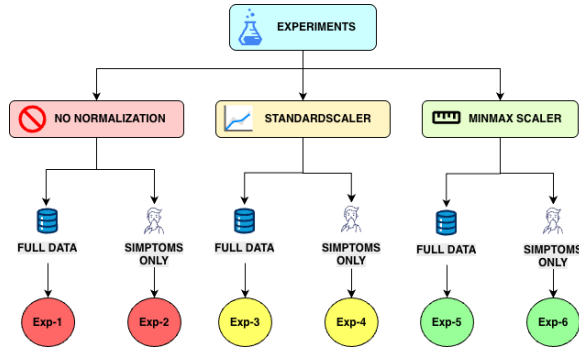


Fig. 2. Normalization Experiment Framework

Fig. 2 illustrates the experimental framework adopted in this study. Two dataset configurations were evaluated: the complete dataset, which includes both clinical symptoms and laboratory examination results, and the symptoms-only dataset, which excludes laboratory features to simulate an early screening scenario. Each dataset configuration was tested using three normalization strategies, namely no normalization, *StandardScaler*, and *MinMaxScaler*, resulting in six experimental scenarios.

StandardScaler standardizes numerical features by transforming them to have zero mean and unit variance,

$$z = \frac{x - \mu}{\sigma}, \quad (1)$$

where μ and σ represent the mean and standard deviation of each feature, respectively [23]. Meanwhile, *MinMaxScaler* rescales feature values into the range of $[0, 1]$ using

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (2)$$

where $x_{\min}$ and $x_{\max}$ denote the minimum and maximum values of each feature [24]. The performance of each experimental scenario was subsequently compared to determine the

most effective normalization strategy for DNN-based dengue fever classification.

TABLE III
NORMALIZATION EXPERIMENT CONFIGURATIONS

Experiment	Dataset	Normalization
Exp-1	Complete Dataset	None
Exp-2	Symptoms Dataset	None
Exp-3	Complete Dataset	StandardScaler
Exp-4	Symptoms Dataset	StandardScaler
Exp-5	Complete Dataset	MinMaxScaler
Exp-6	Symptoms Dataset	MinMaxScaler

E. Proposed Deep Neural Network Model

A Deep Neural Network (DNN) was developed to classify dengue fever cases using clinical and laboratory data. The network architecture consists of an input layer, two fully connected hidden layers, two dropout layers, and a single-neuron output layer, as presented in Table IV. The number of input neurons depends on the selected dataset configuration. ReLU activation was applied in the hidden layers to introduce non-linearity, while dropout layers were incorporated to reduce the risk of overfitting during training. A Sigmoid activation function was used in the output layer to produce binary classification probabilities.

TABLE IV
PROPOSED DNN ARCHITECTURE

Layer	Configuration	Activation
Input Layer	Depends on dataset configuration	–
Dense Layer 1	32 Neurons	ReLU
Dropout Layer 1	Rate = 0.2	–
Dense Layer 2	16 Neurons	ReLU
Dropout Layer 2	Rate = 0.1	–
Output Layer	1 Neuron	Sigmoid

The model was trained using the Adam optimizer with the Binary Crossentropy loss function. Training was performed for a maximum of 50 epochs with a batch size of 32 and an initial learning rate of 0.001. Early stopping with a patience value of five epochs was applied to prevent unnecessary training once the validation performance no longer improved. The complete training configuration is summarized in Table V.

TABLE V
TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Loss Function	Binary Crossentropy
Epochs	50
Batch Size	32
Learning Rate	0.001
Early Stopping	Patience = 5
Cross Validation	5-Fold
Decision Threshold	0.4

A decision threshold of 0.4 was adopted instead of the default value of 0.5 to increase the model's sensitivity in identifying positive dengue cases. From a clinical perspective,

failing to detect an actual dengue patient may have more serious consequences than generating a false positive result, which can be confirmed through subsequent laboratory examinations. All experiments were implemented using TensorFlow 2.x on a workstation equipped with [hardware specification].

F. Evaluation Metrics

The performance of the proposed DNN model was evaluated using several widely adopted classification metrics, including Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) [25], [26]. Accuracy measures the overall proportion of correctly classified instances and is calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

Precision and Recall were used to assess the model's ability to correctly identify positive dengue cases while minimizing false predictions, and are defined as

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN} \quad (4)$$

The F1-Score represents the harmonic mean of Precision and Recall, providing a balanced measure of classification performance, particularly when both false positives and false negatives are important.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

In these equations, *True Positive* (TP) and *True Negative* (TN) denote correctly classified dengue fever and non-dengue fever cases, respectively, whereas *False Positive* (FP) and *False Negative* (FN) represent incorrect classifications. In addition to these metrics, ROC-AUC was used to evaluate the model's discrimination capability across different decision thresholds. Confusion Matrix analysis was also performed to examine the distribution of correct and incorrect predictions and to compare the classification performance obtained from each normalization strategy and dataset configuration.

IV. RESULTS

A. Model Training Performance

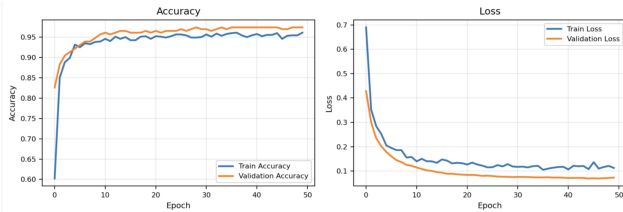


Fig. 3. Training Accuracy and Loss

Fig. 3 presents the learning curves of the proposed DNN model during the training process. Both the training and validation accuracy increased rapidly during the initial epochs

before gradually converging to stable values. The validation accuracy reached approximately 97%, while the training accuracy stabilized at around 95–96%, indicating that the model successfully learned the underlying patterns in the dataset.

A similar trend can be observed in the loss curves, where both training and validation loss decreased consistently as training progressed. The validation loss remained slightly lower than the training loss throughout most epochs, which is likely attributable to the use of dropout regularization during training. Furthermore, the small gap between the training and validation curves suggests that the model generalized well to unseen data and exhibited no significant signs of overfitting.

B. Normalization Experiment Results

Six experimental scenarios were conducted by combining two dataset configurations (complete and symptoms-only) with three normalization approaches: without normalization, *StandardScaler*, and *MinMaxScaler*. Table VI presents an example of how numerical feature values changed after each normalization method.

TABLE VI
COMPARISON OF DATA BEFORE AND AFTER NORMALIZATION

Attribute	Unit	Without Normalization	Standard Scaler	MinMax Scaler
Age	Years	26	0.03	0.33
Temperature	°C	36.7	-1.15	0.18
Systolic BP	mmHg	100	0.05	0.40
Diastolic BP	mmHg	70	0.30	0.25
Platelet	cells/ μ L	139000	-0.33	0.05
Leukocyte	cells/ μ L	4500	-0.64	0.07
Hematocrit	%	26.7	-1.63	0.29
Hemoglobin	g/dL	13.5	-0.30	0.63
Fever Duration	Days	5	1.43	0.71

TABLE VII
NORMALIZATION EXPERIMENT RESULTS

Experiment	Accuracy	Precision	Recall	F1-Score
Exp-1	60.87%	69.23%	39.13%	50.00%
Exp-2	96.96%	95.76%	98.26%	97.00%
Exp-3	95.22%	93.33%	97.39%	95.32%
Exp-4	55.22%	52.78%	99.13%	68.88%
Exp-5	93.04%	93.04%	93.04%	93.04%
Exp-6	93.04%	92.31%	93.91%	93.10%

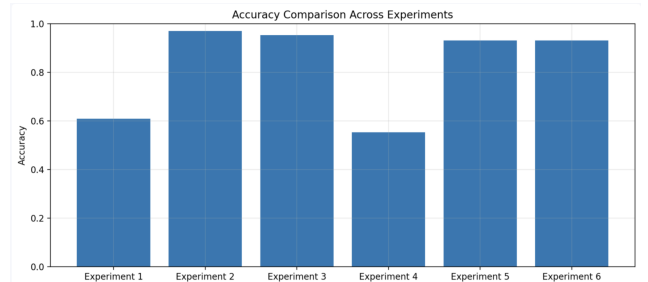


Fig. 4. Accuracy Comparison Across Experiments

Table VII and Fig. 4 demonstrate that feature normalization had a significant impact on model performance. The experiments without normalization (Exp-1 and Exp-4) produced the lowest accuracies, indicating that differences in feature scales hindered the learning process of the DNN model.

Among the evaluated methods, *StandardScaler* achieved the highest performance on the complete dataset (Exp-2), obtaining an accuracy of 96.96% and an F1-score of 97.00%. *MinMaxScaler* also produced competitive results, reaching 95.22% accuracy in Exp-3. For the symptoms-only dataset, both normalization methods achieved consistent accuracies above 93%, substantially outperforming the corresponding experiments without normalization.

Overall, the results indicate that applying feature normalization improves classification performance, with *StandardScaler* providing the best overall results. In addition, the complete dataset consistently outperformed the symptoms-only dataset, suggesting that laboratory features provide valuable information for improving Dengue Fever classification.

C. Classification Evaluation

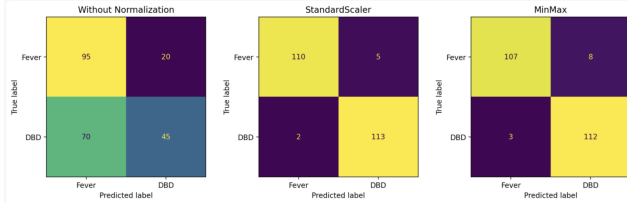


Fig. 5. Confusion Matrix Comparison for Different Normalization Methods

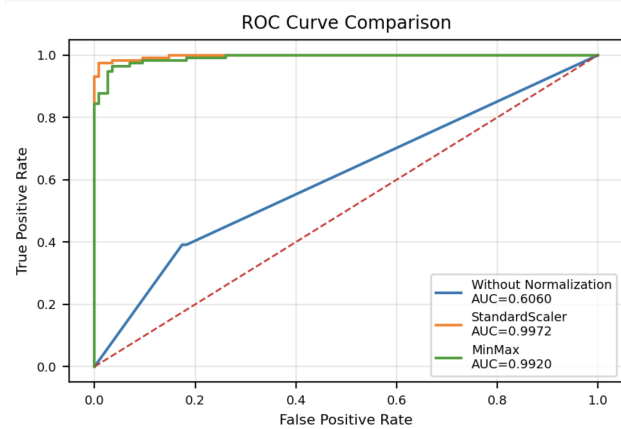


Fig. 6. ROC Curve Comparison for Different Normalization Methods

Figures 5 and 6 present the confusion matrices and ROC curves obtained from the three normalization approaches. The model trained without normalization produced the highest number of misclassified samples, particularly in identifying Dengue Fever cases, indicating that differences in feature scales negatively affected the learning process.

In contrast, both *StandardScaler* and *MinMaxScaler* significantly improved classification performance by reducing

false predictions and increasing the number of correctly classified samples. Among the three approaches, *StandardScaler* achieved the best performance with an AUC of 0.9972, followed closely by *MinMaxScaler* with an AUC of 0.9920. Meanwhile, the model without normalization obtained an AUC of only 0.6060, demonstrating considerably weaker discrimination capability.

Overall, these results indicate that applying feature normalization substantially improves the effectiveness of the proposed DNN model. In particular, *StandardScaler* provides the most reliable classification performance by producing the highest predictive accuracy and the strongest ability to distinguish between Dengue Fever and non-Dengue Fever cases.

V. DISCUSSION

The experimental results demonstrate that data normalization plays an important role in improving the performance of Deep Neural Network (DNN) models for Dengue Fever classification. Among the evaluated methods, *StandardScaler* consistently produced the highest performance across all evaluation metrics. This result indicates that standardizing numerical features to zero mean and unit variance allows the DNN to learn more effectively from clinical data with different value ranges, particularly laboratory measurements such as platelet count, leukocyte count, hematocrit, and hemoglobin. In contrast, models trained without normalization produced substantially lower performance, confirming that differences in feature scales negatively affect the optimization process.

The comparison between dataset configurations also highlights the importance of laboratory examination results. Models trained using the complete dataset consistently outperformed those using only symptom-based features. This finding suggests that laboratory indicators provide additional diagnostic information that complements clinical symptoms, leading to more accurate classification. Nevertheless, the symptoms-only configuration still achieved satisfactory performance above 93%, demonstrating its potential for preliminary screening when laboratory results are not yet available.

The training and validation curves, together with the cross-validation results, indicate that the proposed DNN generalized well to unseen data. The small difference between training and validation performance suggests that the model learned representative patterns without experiencing significant overfitting.

A. Limitations

This study has several limitations. First, the proposed DNN model was not compared with conventional machine learning algorithms such as Support Vector Machine (SVM), Random Forest, or XGBoost, limiting broader performance comparison. Second, the decision threshold of 0.4 was selected empirically to prioritize recall for clinical screening rather than through systematic threshold optimization. Finally, the dataset was obtained from a single data source with a relatively limited number of patient records. Therefore, further validation using larger and more diverse datasets from multiple healthcare

institutions is required to evaluate the robustness and generalizability of the proposed framework.

VI. CONCLUSION

This study investigated the effect of different normalization techniques on the performance of a Deep Neural Network for Dengue Fever classification using clinical symptoms and laboratory data. Experimental results showed that normalization significantly improved classification performance compared to using raw data. Among the evaluated approaches, *StandardScaler* achieved the best overall performance, achieving the highest Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Furthermore, models trained using the full dataset consistently outperformed models using only symptom-based features, indicating that laboratory measurements provide valuable information for disease classification.

The use of nested 5-Fold Cross-Validation, followed by retraining the final model on the full training dataset, also demonstrated stable model performance and good generalization ability. These findings indicate that the proposed framework has strong potential to support early Dengue Fever screening and aid clinical decision-making.

VII. FUTURE WORK

Future research should evaluate the proposed framework using a larger, multi-center clinical dataset to further validate its robustness. Comparison with conventional machine learning algorithms and more sophisticated deep learning architectures would also provide a broader assessment of the model's performance. Furthermore, integrating the proposed model into a real-time, web-based clinical decision support system would facilitate its practical implementation in healthcare settings.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to all parties who contributed to this research, including academic supervisors, healthcare data providers, and supporting institutions.

REFERENCES

- [1] World Health Organization, "Dengue and severe dengue," 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue>
- [2] D. J. Gubler, "Dengue and dengue hemorrhagic fever," *Clinical Microbiology Reviews*, vol. 11, no. 3, pp. 480–496, 1998.
- [3] S. Bhatt *et al.*, "The global distribution and burden of dengue," *Nature*, vol. 496, no. 7446, pp. 504–507, 2013.
- [4] J. Farrar *et al.*, *Dengue: Guidelines for Diagnosis, Treatment, Prevention and Control*. World Health Organization, 2009.
- [5] D. Ravi *et al.*, "Deep learning for health informatics," *IEEE Journal of Biomedical and Health Informatics*, vol. 21, no. 1, pp. 4–21, 2017.
- [6] A. Esteva, B. Kuprel, R. A. Novoa *et al.*, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
- [7] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [8] V. Ramasamy, S. Vadivel *et al.*, "An optimal feature selection with neural network-based classification model for dengue fever prediction," 2023 6th International Conference on Information Systems and Computer Networks (ISCON), pp. 1–5, 2023.
- [9] B. Abdualgalil, S. Abraham, and W. Ismael, "Early diagnosis for dengue disease prediction using efficient machine learning techniques based on clinical data," *Healthcare*, vol. 11, no. 4, 2023.
- [10] S. Garcia *et al.*, "Data preprocessing in data mining," in *Intelligent Systems Reference Library*. Springer, 2015, vol. 72, pp. 1–45.
- [11] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," *International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 1137–1143, 1995.
- [12] D. C. Andrade Girón, W. J. Marín Rodríguez, F. d. M. Lioo-Jordan, and J. L. Ausejo Sánchez, "Machine learning and deep learning models for dengue diagnosis prediction: A systematic review," *Informatics*, 2025.
- [13] H. Hamdani, Z. Arifin, and A. Septiarini, "Expert system of dengue disease using artificial neural network classifier," *JUITA: Jurnal Informatika*, 2022. [Online]. Available: <http://jurnalnasional.ump.ac.id/index.php/JUITA/article/view/12476>
- [14] F. Anibal and other, "Dengue case forecasting using multi-step deep learning models with attention layers," *International Journal of Electrical and Computer Engineering Systems*, 2023.
- [15] V. Ramasamy, S. Vadivel, S. Kothandapani, J. Mahilraj, P. Sivaram, and B. Sharma, "An optimal feature selection with neural network-based classification model for dengue fever prediction," pp. 1–5, 2023.
- [16] T. Xu and G. Xiao, "Deep learning-based predictive modeling of dengue fever incidence in southeast asia," *International Journal of Infectious Diseases*, vol. 167, p. 108592, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1201971226002274>
- [17] D. Ravi, C. Wong, F. Deligianni *et al.*, "Deep learning for health informatics," *IEEE Journal of Biomedical and Health Informatics*, vol. 21, no. 1, pp. 4–21, 2017.
- [18] Hamdani and other, "Dengue classification method using support vector machines and cross-validation techniques," *IAES International Journal of Artificial Intelligence*, 2022.
- [19] V. Ramasamy, S. Vadivel, S. Kothandapani, J. Mahilraj, S. Ponnusamy, and B. Sharma, "An optimal feature selection with neural network-based classification model for dengue fever prediction," pp. 1–5, 03 2023.
- [20] B. Behera, G. Kumaravelan, and P. Borugadda, "Performance evaluation of deep learning algorithms in biomedical document classification," pp. 220–224, 12 2019.
- [21] B. Abdualgalil, S. Abraham, and W. M. Ismael, "Early diagnosis for dengue disease prediction using efficient machine learning techniques based on clinical data," *Healthcare*, 2023. [Online]. Available: <https://journal.ummy.ac.id/index.php/jrc/article/view/14387>
- [22] F. Cholet, *Deep Learning with Python for Medical Data Classification*. Manning Publications, 2017.
- [23] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. O'Reilly Media, 2022.
- [24] S. learn Developers, "Minmaxscaler documentation," <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>, 2024, accessed: 2026-05-30.
- [25] D. M. W. Powers, "Evaluation: From precision, recall and f-measure to roc, informedness, markedness and correlation," *Journal of Machine Learning Technologies*, vol. 2, no. 1, pp. 37–63, 2011.
- [26] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," *Information Processing and Management*, vol. 45, no. 4, pp. 427–437, 2009.